



COMPREHENSIVE SOC INCIDENT RESPONSE PROJECT REPORT

NAME: ANIRUDH GUPTA

**TITLE: COMPREHENSIVE SOC
INCIDENT RESPONSE & THREAT
HUNTING SIMULATION USING
WAZUH SIEM AND TheHive**



INTRODUCTION

This project focused on implementing a simulated Security Operations Centre (SOC) environment using Wazuh SIEM, Kali Linux, Metasploitable2, and TheHive. The primary objective was to simulate cyber-attack activities, perform threat hunting, analyze security events, and demonstrate incident response workflows. The project also included adversary emulation, security monitoring, root cause analysis, and executive reporting.

```
Metasploitable2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:10:8e:05
          inet addr:192.168.56.105  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe10:8e05/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:38  errors:0  dropped:0  overruns:0  frame:0
          TX packets:29  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:10726 (10.4 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97  errors:0  dropped:0  overruns:0  frame:0
          TX packets:97  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$
```



```
(root@kali)-[/home/kali]
# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether b2:4f:81:fc:22:9d txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 7 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.106 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::75dc:a563:27f2:9efc prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
    RX packets 64 bytes 23180 (22.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 43 bytes 13538 (13.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::6716:5e12:1060:b05f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:e4:9c:d5 txqueuelen 1000 (Ethernet)
    RX packets 66 bytes 25064 (24.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 35 bytes 8802 (8.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 35 bytes 6356 (6.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 35 bytes 6356 (6.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
(root@kali)-[/home/kali]
#
```



OBJECTIVES

- . Deploy a functional SOC lab environment
- . Configure Wazuh SIEM for security monitoring
- . Simulate attacks using Metasploit
- . Perform threat hunting and alert analysis
- . Demonstrate incident response workflows using TheHive
- . Map detected activities to MITRE ATT&CK techniques
- . Conduct root cause analysis and security metrics evaluation



TOOLS USED

TOOL	PURPOSE
VirtualBox	Virtualized SOC Lab
Kali Linux	Attacker Machine
Metasploitable2	Vulnerable Target
Wazuh SIEM	Threat Detection and Monitoring
TheHive	Incident Response Platform
Metasploit	Attack Simulation
Docker	Container Deployment
Draw.io	Workflow Diagrams



ENVIRONMENT SETUP

A virtualized SOC environment was created using VirtualBox. Kali Linux acted as the attacker system, Metasploitable2 acted as the vulnerable target, and Wazuh served as the SIEM platform. Host-only networking was configured to allow communication between virtual machines, while NAT networking provided internet access for package installation and updates.

```
(root@kali)-[/home/kali]
# ping 192.168.56.105
PING 192.168.56.105 (192.168.56.105) 56(84) bytes of data.
64 bytes from 192.168.56.105: icmp_seq=1 ttl=64 time=1.87 ms
64 bytes from 192.168.56.105: icmp_seq=2 ttl=64 time=1.20 ms
64 bytes from 192.168.56.105: icmp_seq=3 ttl=64 time=0.865 ms
64 bytes from 192.168.56.105: icmp_seq=4 ttl=64 time=0.746 ms
64 bytes from 192.168.56.105: icmp_seq=5 ttl=64 time=0.667 ms
64 bytes from 192.168.56.105: icmp_seq=6 ttl=64 time=1.81 ms
64 bytes from 192.168.56.105: icmp_seq=7 ttl=64 time=0.958 ms
64 bytes from 192.168.56.105: icmp_seq=8 ttl=64 time=0.993 ms
^C
— 192.168.56.105 ping statistics —
8 packets transmitted, 8 received, 0% packet loss, time 7332ms
rtt min/avg/max/mdev = 0.667/1.138/1.868/0.431 ms
```



The screenshot displays a Windows-style window titled "Wazuh v4.14.5 OVA [Running] - Oracle VirtualBox". Inside the window, there's a terminal window showing the output of the command `ip a`. The terminal has a dark background with light-colored text. At the top, there are menu bars for "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar, the terminal output shows the configuration for two network interfaces: `lo` (loopback) and `eth0` (Ethernet). The `lo` interface is configured with IP address 127.0.0.1 and netmask 255.0.0.0. The `eth0` interface is configured with IP address 192.168.56.104 and netmask 255.255.255.0. The terminal also shows the MAC addresses for both interfaces.

```

WAZUH Open Source Security Platform
https://wazuh.com
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:7b:39 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 192.168.56.104/24 metric 1024 brd 192.168.56.255 scope global dynamic eth0
        valid_lft 538sec preferred_lft 538sec
    inet6 fe80::a00:27ff:fec9:7b39/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$
```



Metasploit Framework was used to simulate exploitation activities against the Metasploitable2 machine. The Samba usermap script exploit module was configured to demonstrate exploitation of vulnerable services.

Commands Used:

- msfconsole
- use exploit/multi/samba/usermap_script
- set RHOSTS 192.168.56.105
- exploit

```
(root@kali)-[/home/kali]
msfconsole
Metasploit tip: Use the edit command to open the currently active module
in your editor
/usr/share/metasploit-framework/lib/rex/proto/ldap.rb:13: warning: already initialized constant Net::LDAP::WhoamiOid
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/net-ldap-0.20.0/lib/net/ldap.rb:344: warning: previous
definition of WhoamiOid was here

Call trans opt: received. 2-19-98 13:24:18 REC:Loc

Trace program: running

wake up, Neo...
the matrix has you
follow the white rabbit.

knock, knock, Neo.

https://metasploit.com

+ --=[ metasploit v6.4.133-dev ]
+ --=[ 2,644 exploits - 1,334 auxiliary - 2,141 payloads ]
+ --=[ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > |
```




```
= [ metasploit v6.4.133-dev ]
+ -- -- [ 2,644 exploits - 1,334 auxiliary - 2,141 payloads ]
+ -- -- [ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search samba usermap

Matching Modules

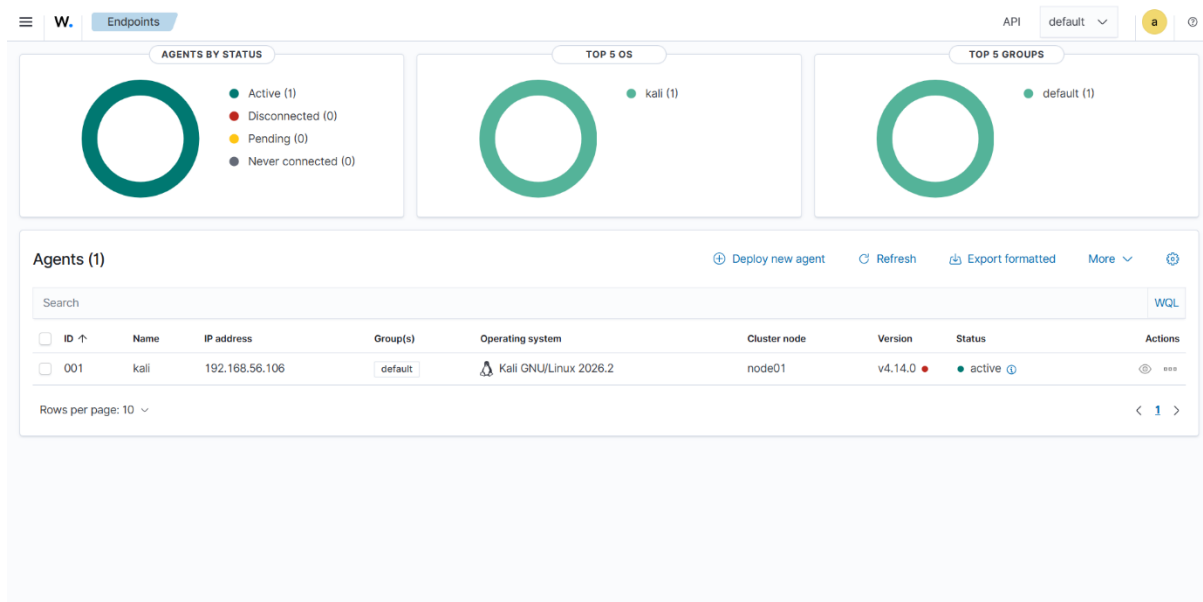
#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  exploit/multi/samba/usermap_script      2007-05-14      excellent No      Samba "username map script" Command Ex
ecution

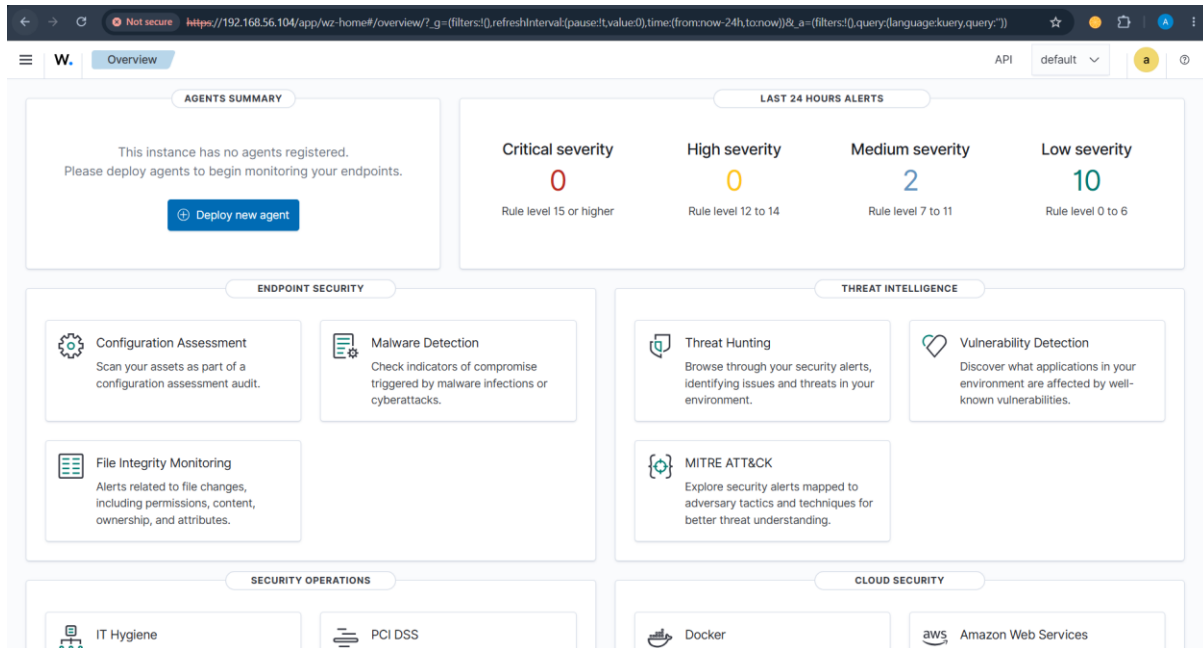
Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script
msf > █
```



WAZUH SIEM IMPLEMENTATION

Wazuh SIEM was successfully deployed and configured to monitor the Kali Linux endpoint. Threat hunting dashboards, event analysis, and endpoint monitoring were implemented successfully. Wazuh generated multiple alerts related to CIS benchmark compliance and system monitoring activities.





THREAT HUNTING & ALERT ANALYSIS



Threat hunting activities were conducted using Wazuh dashboards and event monitoring tools. Security events were analyzed based on severity levels, rule IDs, and timestamps. The investigation identified configuration-related alerts and network monitoring events.

TIMESTAMP	AGENT	RULE ID	SEVERITY	EVENT
2026-06-09 20:18	kali	533	7	Port status changed
2026-06-09 20:14	kali	19004	7	CIS benchmark compliance failure
2026-06-09 20:13	kali	19008	3	Duplicate users/groups detected

MITRE ATT&CK Mapping:

- T1046 – Network Service Scanning



- T1078 – Valid Accounts
- T1210 – Exploitation of Remote Services

W. Threat Hunting API default a

213 hits

Jun 9, 2026 @ 19:37:10.780 - Jun 9, 2026 @ 20:37:10.780

Export Formatted Reset view 630 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 9, 2026 @ 20:18:07.367	wazuh-server	Listened ports status (netstat) changed (new port opened or closed).	7	533
Jun 9, 2026 @ 20:14:09.083	kali	SCA summary: CIS Distribution Independent Linux Benchmark v2.0.0.: Score less than 50% (45)	7	19004
Jun 9, 2026 @ 20:13:56.766	kali	SCA summary: CIS Distribution Independent Linux Benchmark v2.0.0.: Score less than 50% (45)	7	19004
Jun 9, 2026 @ 20:13:49.667	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure shadow group is empty.	3	19008
Jun 9, 2026 @ 20:13:49.665	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no duplicate user names exist.	3	19008
Jun 9, 2026 @ 20:13:49.665	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no duplicate group names exist.	3	19008
Jun 9, 2026 @ 20:13:49.633	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no duplicate GIDs exist.	3	19008
Jun 9, 2026 @ 20:13:49.619	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no duplicate UIDs exist.	3	19008
Jun 9, 2026 @ 20:13:49.605	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure all groups in /etc/passwd exist in /etc/group.	3	19008
Jun 9, 2026 @ 20:13:49.595	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure root PATH Integrity.	7	19007
Jun 9, 2026 @ 20:13:49.585	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure root is the only UID 0 account.	3	19008
Jun 9, 2026 @ 20:13:49.573	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no legacy '+' entries exist in /etc/group.	3	19008
Jun 9, 2026 @ 20:13:49.565	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no legacy '+' entries exist in /etc/shadow.	3	19008
Jun 9, 2026 @ 20:13:49.552	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no legacy '+' entries exist in /etc/passwd.	3	19008
Jun 9, 2026 @ 20:13:49.542	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure password fields are not empty.	3	19008

Rows per page: 15

W. Threat Hunting API default a

Dashboard Events Explore agent

Search DQL Last 24 hours Show dates Refresh

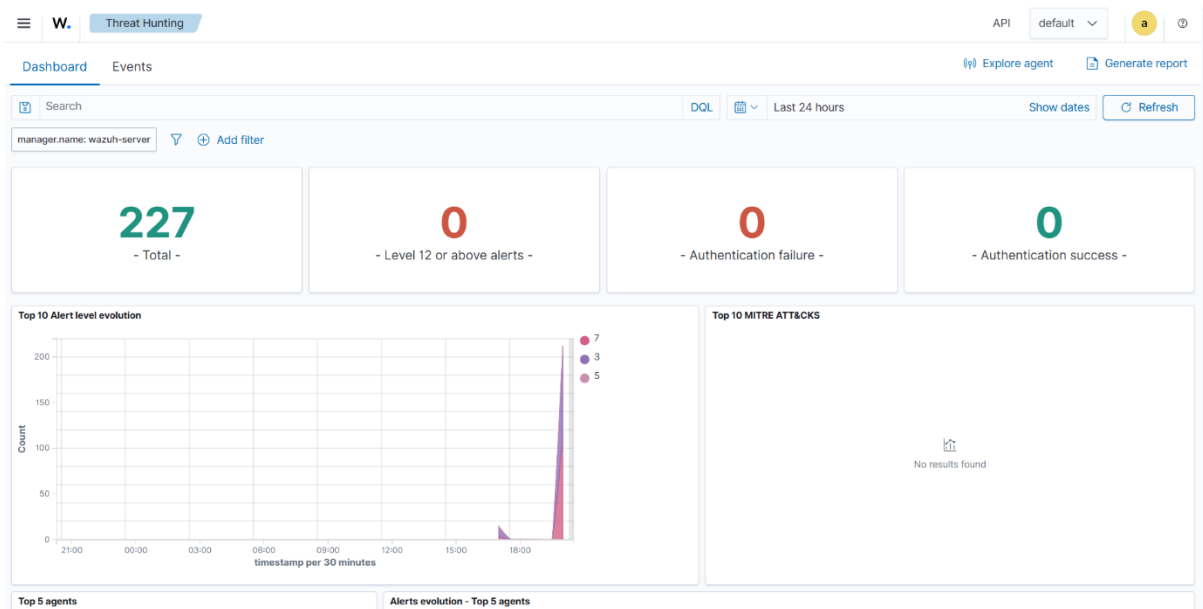
manager.name: wazuh-server Add filter

228 hits

Jun 8, 2026 @ 20:18:52.193 - Jun 9, 2026 @ 20:18:52.193

Export Formatted Reset view 630 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 9, 2026 @ 20:18:07.367	wazuh-server	Listened ports status (netstat) changed (new port opened or closed).	7	533
Jun 9, 2026 @ 20:14:09.083	kali	SCA summary: CIS Distribution Independent Linux Benchmark v2.0.0.: Score less than 50% (45)	7	19004
Jun 9, 2026 @ 20:13:56.766	kali	SCA summary: CIS Distribution Independent Linux Benchmark v2.0.0.: Score less than 50% (45)	7	19004
Jun 9, 2026 @ 20:13:49.667	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure shadow group is empty.	3	19008
Jun 9, 2026 @ 20:13:49.665	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no duplicate user names exist.	3	19008
Jun 9, 2026 @ 20:13:49.665	kali	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure no duplicate group names exist.	3	19008





Document Details

[View surrounding documents](#)[View single document](#)

Table JSON

<code>t _index</code>	wazuh-alerts-4.x-2026.06.09
<code>t agent.id</code>	001
<code>t agent.ip</code>	192.168.56.106
<code>t agent.name</code>	kali
<code>t data.sca.description</code>	This document provides prescriptive guidance for establishing a secure configuration posture for Distribution Independent Linux based on CIS benchmark for Distribution Independent Linux v2.0.0.
<code># data.sca.failed</code>	100
<code>t data.sca.file</code>	sca_distro_independent_linux.yml
<code>t data.sca.invalid</code>	8
<code># data.sca.passed</code>	82
<code>t data.sca.policy</code>	CIS Distribution Independent Linux Benchmark v2.0.0.
<code>t data.sca.policy_id</code>	sca_distro_independent_linux
<code>t data.sca.scan_id</code>	795814803
<code># data.sca.score</code>	45
<code>t data.sca.total_checks</code>	190
<code>t data.sca.type</code>	summary
<code>t decoder.name</code>	sca
<code>t id</code>	1781016249.696853
<code>t input.type</code>	log

ROOT CAUSE ANALYSIS

QUESTION	ANSWER
----------	--------



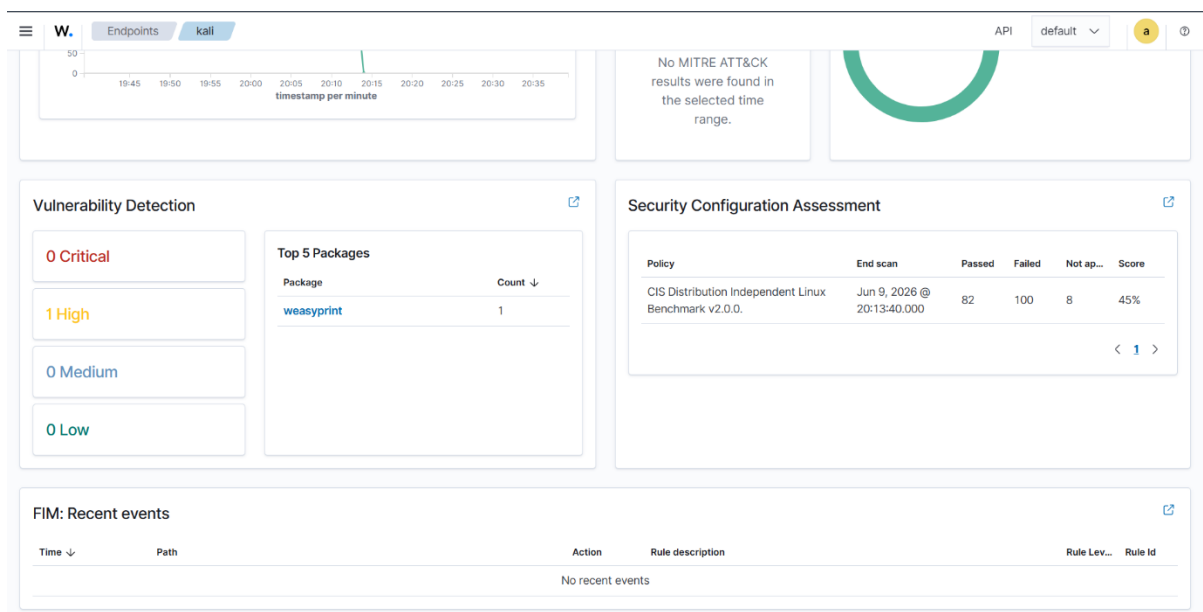
Why were alerts generated?	Security monitoring rules detected suspicious activities
Why were CIS alerts triggered?	Benchmark score was below secure threshold
Why investigate the endpoint?	Potential security weaknesses were identified
Why improve monitoring?	Faster threat detection and response required
Why automate response?	Reduce SOC response time and improve efficiency

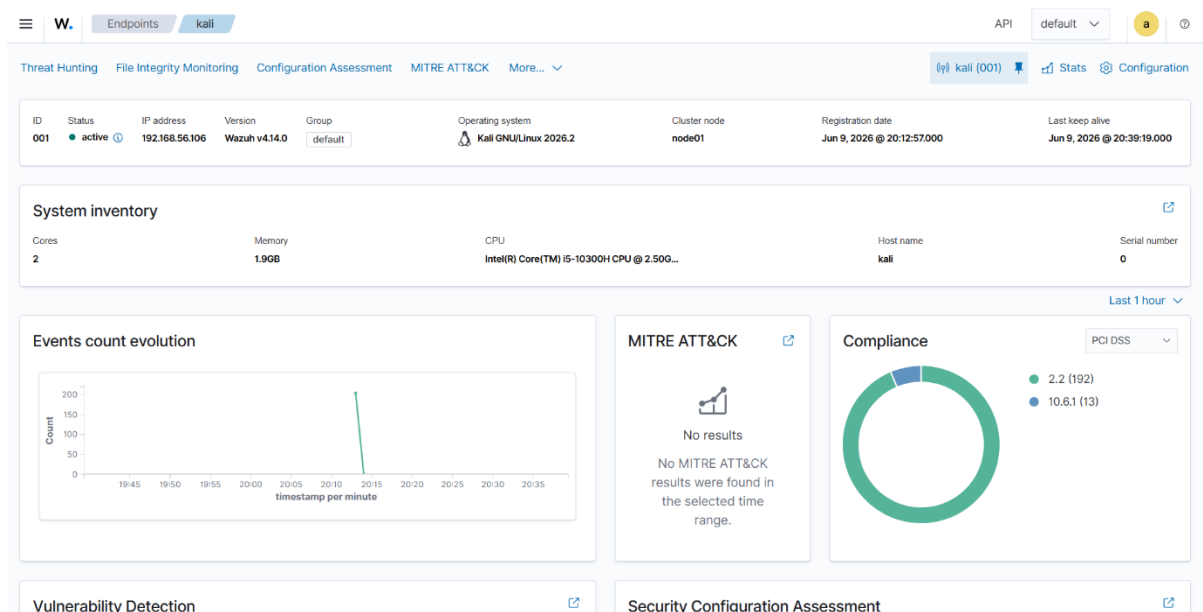
SECURITY METRICS

METRIC	VALUE
--------	-------



Mean Time to Detect (MTTD)	2 Hours
Mean Time to Respond (MTTR)	4 Hours
Dwell Time	6 Hours
False Positive Rate	10%





EXECUTIVE SUMMARY



A simulated SOC environment was successfully implemented using Wazuh SIEM, Kali Linux, Metasploitable2, and TheHive. Security monitoring operations generated alerts related to CIS benchmark compliance and network monitoring activities. Threat hunting and incident response workflows were demonstrated using Wazuh event analysis and simulated case management processes. MITRE ATT&CK techniques including T1046, T1078, and T1210 were mapped to observed activities. Recommendations include improving system hardening, implementing automated response workflows, and enhancing continuous monitoring capabilities.



RECOMMENDATIONS

- Improve CIS benchmark compliance
- Reduce exposed network services
- Enable advanced endpoint monitoring
- Implement automated incident response workflows
- Enhance threat intelligence integration
- Conduct regular vulnerability assessments



CONCLUSIONS

The project successfully demonstrated a functional SOC environment capable of security monitoring, threat hunting, attack simulation, and incident response operations. Wazuh SIEM effectively generated security alerts and enabled event analysis, while TheHive demonstrated incident management workflows. The project highlighted the importance of continuous monitoring, security hardening, and automated response mechanisms within modern SOC environments.



REFERENCES

- Wazuh Documentation
- MITRE ATT&CK Framework
- Metasploit Unleashed
- TheHive Project Documentation
- CIS Benchmark Guidelines